# Quiz 11

**Friday, December 4 — eigenvalues, complex numbers, and Lab 5 (Week 13)**

Last updated: October 12, 2026.

**Textbook:** Boyd & Vandenberghe, *VMLS* (PDF)

**Strang / MIT (optional):** Reading: Strang §6.1 (PDF) — eigenvalues
Video: 18.06 Lec 21 — eigenvalues and eigenvectors · MIT 18.06 playlist (Strang)

**Covers the prior instructional week only:** Eigenvectors, Characteristic, Eigenbehavior, and Lab 5 (Week 13). Fall break is skipped when choosing the prior week. Same-day SVD / Lab 6 / least squares (Week 15) are **not** on this quiz.

Textbooks: Strang §6.1 · Lachniet Octave PDF · also on Resources.

Schedule

**SVD and least squares are not on this quiz**; see Lab 6, the capstone packet, and the final.

# Core

## Fluency (complete prompts — practice these by hand)

**C11.1.** Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Compute $Av$. Is $v$ an eigenvector? If so, what is $\lambda$?

**C11.2.** You are told that $v = (1, -1)$ is an eigenvector of

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

Find $\lambda$ by comparing components of $Av$ and $v$. Then verify $Av = \lambda v$.

**C11.3.** Let $A = \mathrm{diag}(1.2,\ 0.5)$ (Lecture 28 / Lab 5 long-run pattern).

(a) Name two real eigenpairs using the standard basis.
(b) If $x_0 = (1, 1)$, write $x_t = A^t x_0$ componentwise. Which term dominates for large $t$?

**C11.4.** Complex drill. Let $z = 2 + i$ and $w = 1 - 3i$. Compute $z + w$, $zw$, and $|z|$.

**C11.5.** For

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix},$$

form $\det(A - \lambda I)$, find both eigenvalues, and give one eigenvector for $\lambda = 3$.

**C11.6.** Rotation. Let $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (90° CCW).

(a) Show that the characteristic polynomial is $\lambda^2 + 1 = 0$, so $\lambda = \pm i$.
(b) Why does this match the geometric fact that $R$ has no real eigenvector?

**C11.7.** Check a complex pair. With $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $v = (1, -i)$, compute $Av$ and decide whether $Av = iv$ or $Av = (-i)v$.

**C11.8.** True or false, with a one-line reason: if $Av = 2v$ and $v \neq 0$, then $A^3 v = 8v$.

**C11.9 (Lab 5 / software).** Suppose Octave/Python reports eigenvalues of a $2 \times 2$ matrix as approximately 0.9 and 0.2, with corresponding eigenvectors $v_1 \approx (0.7,\ 0.7)$ and $v_2 \approx (-0.7,\ 0.7)$ (normalized). For dynamics $x_{t+1} = Ax_t$ started near $v_1 + v_2$, which mode dominates for large $t$, and does the state grow or decay?

**C11.10 (Lab 5 / verify).** Software prints $\lambda \approx 3$ and $v \approx (0.707, 0.707)$ for

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Explain how you would verify $Av \approx \lambda v$ numerically in one or two lines of code (Octave or Python). What should $\|Av - \lambda v\|$ look like if the pair is correct?

### Textbook Core

- S 6.1 #1, #2, #5, #8, #10, #12, #14, #16, #17
- L Ch. 5 Ex. 1: selected hand/software eigenvalue parts
- L Ch. 5 Ex. 2: Markov equilibrium/eigenvector portions
- L Ch. 5 Ex. 5: selected matrix-power portion

## Stretch

S 6.1 #3 and #4; a complex eigenpair check beyond C11.7.

## What a quiz item might change

A new $2 \times 2$ matrix with a factorable characteristic polynomial, a proposed eigenpair to verify, a Markov transition probability, a rotation matrix with complex eigenvalues, or a software printout to interpret (dominant $\lambda$, verify $Av \approx \lambda v$).